Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/28/2006
PI Name:	Sayler, Gary Ph.D.		
Project Title:	Bioluminescent Monitoring of Opportunistic Pathogens in the Spacecraft Environment		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sayler@utk.edu	**Fax:**	FY 865-974-8086
PI Organization Type:	UNIVERSITY	**Phone:**	865-974-8080
Organization Name:	University of Tennessee		
PI Address 1:	676 Dabney Hall		
PI Address 2:			
PI Web Page:	http://www.ceb.utk.edu		
City:	Knoxville	**State:**	TN
Zip Code:	37996	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-01
Start Date:	12/31/2002	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ripp, Steven (The University of Tennessee) Simpson, Michael (Oak Ridge National Laboratory) Garland, Jay L (Dynamac Corp., Kennedy Space Center)		
Grant/Contract No.:	NAG9-1424		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The specific research objective of this proposal is to insert a quorum sensing gene (luxI) into bacteriophage to intrinsically generate autoinducer molecules upon target cell infection and subsequent amplification of lux-based bioluminescent signaling mechanisms. Using quorum sensing molecules as a means of detection allows for sensing of biological agents at very low level concentrations. As opposed to chemical detection systems, where typical toxic thresholds of part-per-billion to part-per-million still equate to >1014 molecules available for detection, a quorum sensing biological detection system may be capable of sensing down to one individual biological agent. This technology provides a unique mechanism to provide a measurable amount of bioluminescence from these low event numbers by using the production of autoinducer molecules to produce amplification of individual targeted biological agents, thus avoiding dependence upon target cell growth.		

	To develop microelectronic-based bacteriophage-amplified bioluminescent biosensors for the specific detection and identification of biological contaminants in the spacecraft environment. The sensors are capable of functioning as a network of autonomous detectors for continuous on-line, near real-time monitoring in a low-power, low-mass, low-maintenance, redundant framework.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ability to rapidly detect target pathogens has earth benefits in medical diagnostics, food safety, environmental monitoring, and homeland defense.
Task Progress:	Our testing of these reporter phage systems indicates detection limits approaching 10e4 CFU/ml without sample preincubation and down to 10 CFU/ml with sample preincubation for E. coli O157:H7 within a 96-well plate assay format. Testing in artificially contaminated apple juice demonstrated detection limits down to 10e4 CFU/ml in less than 8 hours. For general E. coli detection using strain XL1-Blue as a target, identification in pure culture occurred within 1.5 – 5 hours at high cell densities (10e8 – 10e5 CFU/ml) and at lower cell densities (<10 CFU/ml) with requisite preincubation within a total assay time of approximately 10 hours. When tested against artificially contaminated lettuce, the assay detected original inoculums between 10e8 – 10e5 CFU/ml within 2.6 – 12 hours, respectively. Optimization of Salmonella detection assays is still ongoing. Due to the unexpected ability of recombinant reporter phage P22 to infect but not lyse the OHHLux bioreporter cells, the assay will require reengineering of the OHHL responsive bioluminescent bioreporter into a noninfectable strain. We have chosen to do so in Pseudomonas aeruginosa (ATCC BAA-47). This should be a simple matter of removing the plasmid from OHHLux and placing it into P. aeruginosa. Once complete, we will have a new bioluminescent bioreporter to use in the assay that is resistant to P22 infection. While the overall assay limits fall short of the 1 – 10 CFU/ml detection limits typically desired in food matrices, their limits of detection are close to that of existing immunoassays and fluorescent-phage based assays (5), and the sensitivity of these assays may be increased through additional genetic modifications of the phage (i.e., linkage of luxI to multiple phage promoters) or through better sample concentration techniques (i.e., immunomagnetic separation). Since no ancillary reagents or manipulations are required other than incubation and light measurement, these assays remain highly amenable to automated high-throughput monitoring using any number of available photomultiplier-based luminescent readers, including miniaturization in a BBIC format. Phage-based sensing also ensures only the detection of live cells, thereby overcoming the false positive detection of dead cells often inherent in other assaying methods such as ELISA and PCR. Further manipulations of phage reporter genetic mechanisms and advances in miniaturized electronic optics will clearly impact future monitoring and detection strategies in the food industry.
Bibliography Type:	Description: (Last Updated: 03/16/2017)
Abstracts for Journals and Proceedings	Birmele M, Ripp S, Sayler G, Jegier P, Garland J. "Development and Validation of the Bioluminescent Monitoring System for Escherichia coli in the Environment." American Society for Microbiology Annual Meeting, Atlanta, GA, March 2005. American Society for Microbiology Annual Meeting, proceedings, 2005. , Mar-2005
Abstracts for Journals and Proceedings	Iakova P, Ripp S, Garland J, Sayler G. "Binary phage amplified bioluminescent sensing of Salmonella." American Society for Microbiology Annual Meeting, Atlanta, GA, March 2005. American Society for Microbiology Annual Meeting, proceedings, 2005. , Mar-2005
Abstracts for Journals and Proceedings	Sayler G, Ripp S, Melechko A, Mitchel M. "Bioluminescent bioreporter integrated circuit for phage amplified bacterial detection." International Union of Microbiological Societies, San Francisco, CA, July 2005. International Union of Microbiological Societies, proceedings, July 2005. , Jul-2005
Articles in Other Journals or Periodicals	Ripp S. "Biosensor monitoring of the cleanroom environment." CleanRooms Magazine. 2006 Apr 20(4):25-9. , Apr-2006
Articles in Peer-reviewed Journals	Brigati JR, Ripp SA, Johnson CM, Iakova PA, Jegier P, Sayler GS. "Bacteriophage-based bioluminescent bioreporter for the detection of Escherichia coli O157:H7." Journal of Food Protection, 2007 Jun;70(6):1386-92. PMID: 17612068 , Jun-2007
Articles in Peer-reviewed Journals	Vijayaraghavan R, Islam SK, Zhang M, Ripp S, Caylor S, Bull N, Moser S, Terry S, Blalock B, Sayler GS. "A bioreporter bioluminescent integrated circuit for very low-level chemical sensing in both gas and liquid environments." Sens Actuators B Chem, 2007 May 21;123(2):922-8. http://dx.doi.org/10.1016/j.snb.2006.10.064 , May-2007
Articles in Peer-reviewed Journals	Islam SK, Vijayaraghavan R, Zhang M, Ripp S, Caylor S, Weathers B, Moser S, Terry S, Blalock B, Sayler GS. "Integrated circuit biosensors using living whole-cell bioreporters." IEEE Transactions on Circuits and Systems--I: Regular Papers, 2007 Jan;54(1):89-98. http://dx.doi.org/10.1109/TCSI.2006.887982 , Jan-2007
Articles in Peer-reviewed Journals	Ripp S, Jegier P, Birmele M, Johnson CM, Daumer KA, Garland JL, Sayler GS. "Linking bacteriophage infection to quorum sensing signalling and bioluminescent bioreporter monitoring for direct detection of bacterial agents." J Appl Microbiol. 2006 Mar;100(3):488-99. PMID: 16478488 , Mar-2006
Articles in Peer-reviewed Journals	Ripp S, Jegier P, Johnson CM, Brigati JR, Sayler GS. "Bacteriophage-amplified bioluminescent sensing of Escherichia coli O157:H7." Anal Bioanal Chem. 2008 Jan 10; [Epub ahead of print] PMID: 18188543 , Jan-2008

Papers from Meeting Proceedings	Ripp S, Garland JL, Blalock BJ, Islam SK, Sayler GS. "Bioluminescent bioreporter integrated circuit sensing of the chemical and biological spacecraft environment." Bioluminescence & chemiluminescence : progress and perspectives. Proceedings of the 13th International Symposium on Bioluminescence and Chemiluminescence held at Pacifico Yokohama, Yokohama, Japan from 2 - 6 August 2004. Ed. by A. Tsuji, et al. In: Bioluminescence & chemiluminescence : progress and perspectives. River Edge, N.J. : World Scientific, c2005. p. 433-436. , Jan-2005
Presentation	Ripp, S. "Bioluminescent bioreporter sensing of the chemical and biological environment" Environmental Sentinels 2005 Jun-2005
Presentation	Ripp, S. "Biophotonic biosensors" Office of Naval Research Interagency Consortium Jun-2005